

## Seewoo Lee

Ph. D. student in Mathematics

University of California Berkeley, US

seewoo5@berkeley.edu

🏠 <https://seewoo5.github.io>

🐙 <https://github.com/seewoo5>

🌐 <https://www.linkedin.com/in/lee-seewoo-857062171>

### Education

#### University of California Berkeley

Ph.D in Mathematics

Berkeley

2018 – 2026

- On leave for military service (2019 Fall - 2022 Summer)
- Advisor: Sug Woo Shin

#### Pohang University of Science and Technology (POSTECH)

M.S in Mathematics

Pohang

2017 – 2018

- Thesis: *Maass wave forms, quantum modular forms and Hecke operators*
- Advisor: YoungJu Choie

#### Pohang University of Science and Technology (POSTECH)

B.S. in Mathematics

Pohang

2013 – 2017

- *Summa Cum Laude* with top honours in mathematics
- Honor's thesis: *Quantum modular forms and Hecke operators*

### Experiences

#### Axiom Math

Internship

Palo Alto

2026.01 – 2026.04

- Machine Learning & AI for Mathematical Discovery and Reasoning

#### CryptoLab

Research Engineer

Seoul

2021.05 – 2022.07

- Research on Homomorphic Encryption and application in Machine Learning

#### Riidx!

Research Scientist

Seoul

2019.07 – 2021.05

- Research on Knowledge Tracing, Score Prediction, Student Dropout Prediction, Item Recommendation

### Research Interests

- Automorphic Forms and Representations, Computational Number Theory, Relative Langlands Program
- Machine Learning and Deep Learning, Formalization of Mathematics, Homomorphic Encryption

## Publications

- Math

1. **S. Lee**, *Algebraic proof of modular form inequalities for optimal sphere packings*. To appear in Algebra and Number Theory, arXiv:2406.14659
2. D. Angdinata, E. Chen, **S. Lee**, K. Ono, J. Zhang, *ABC implies that Ramanujan's tau function misses almost all primes*, To appear in Indagationes Mathematicae, arXiv:2603.29970
3. E. Chen, K. Lau, **S. Lee**, K. Ono, J. Zhang, *Almost all primes are partially regular*, To appear in Archiv der Mathematik, arXiv:2602.05090
4. K. Lee, **S. Lee**, *Machines Learn Number Fields, But How? The Case of Galois Groups*, To appear in Research in Mathematical Sciences, arXiv:2508.06670
5. **S. Lee**, *Shanks bias in function fields*, To appear in Journal de Théorie des Nombres de Bordeaux, arXiv:2509.16142
6. J. Baek, **S. Lee**, *An equilateral triangle of side  $> n$  cannot be covered by  $n^2 + 1$  unit equilateral triangles homothetic to it*, American Mathematical Monthly, 1-9 (2024)
7. D. Choi, **S. Lee**, *Non-archimedean Sendov's conjecture,  $p$ -adic numbers*, Ultrametric Analysis and Applications 14, 77-80 (2022)
8. **S. Lee**, *Maass wave forms, Quantum Modular Forms and Hecke Operators*, Res. Mathematical Science 6, 7 (2018), Modular Forms are Everywhere: Celebration of Don Zagier's 65th Birthday
9. **S. Lee**, *Quantum Modular Forms and Hecke Operators*, Res. Number Theory 4, 18 (2018)
10. Y. Chen, R. Chernov, M. Flores, M. F. Bourque, **S. Lee**, B. Yang, *Toy Teichmüller spaces of real dimension 2: the pentagon and the punctured triangle*, Geom. Dedicata 197 (2018), 193-227

- Others

1. F. Lin, K. Nagel, **S. Lee**, J. Jiang, G. Yang, P. Chang, S. Li, N. Sheu, *An Analysis of Silk Density in Spider Webs*, Royal Society Open Science 12: 250455 (2025)
2. **S. Lee**, G. Lee, J. Kim, J. Shin, M. Lee, *HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption*, International Conference on Machine Learning. 2023 (Oral, 155/6538)
3. **S. Lee**, J. Kim, *Revisiting the Convergence Theorem for Competitive Bidding in Common Value Actions*, Economic Theory Bulletin 10, 293-302 (2022)
4. **S. Lee**, K. Kim, J. Shin, J. Park, *Tracing Knowledge for Tracing Dropouts: Multi-Task Training for Study Session Dropout Prediction*, Educational Data Mining. 2021
5. M. Kim, Y. Shim, **S. Lee**, H. Loh, J. Park, *Behavioral Testing of Deep Knowledge Tracing Models*, Educational Data Mining 2021
6. H. Loh, D. Shin, **S. Lee**, J. Baek, C. Hwang, Y. Lee, Y. Cha, S. Kwon, J. Park and Y. Choi, *Recommendation for Effective Standardized Exam Preparation*, LAK21: 11th International Learning Analytics and Knowledge Conference. 2021
7. D. Shin, Y. Shim, H. Yu, **S. Lee**, B. Kim, Y. Choi, *SAINT+: Integrating Temporal Features for EdNet Correctness Prediction*, LAK21: 11th International Learning Analytics and Knowledge Conference. 2021
8. Y. Choi, Y. Lee, D. Shin, J. Cho, S. Park, **S. Lee**, J. Baek, C. Bae, B. Kim, J. Heo, *EdNet: A Large-Scale Hierarchical Dataset in Education*, International Conference on Artificial Intelligence in Education (2021), 69-73
9. J. Kim, **S. Lee**, *Joint Liability and Stochastic Shapley Value*, International Review of Law & Economics 60 (2019), 1-8

## Preprints

1. S. Hariharan, C. Birkbeck, **S. Lee**, G. Ma, B. Mehta, A. Poiroux, M. Viazovska, *A Milestone in Formalization: The Sphere Packing Problem in Dimension 8*, arXiv:2604.23468, submitted
2. G. Bates, R. Jesubalan, **S. Lee**, J. Lu, H. Shim, *Ties in function field prime race*, arXiv:2603.21005, submitted
3. **S. Lee**, *Inequalities involving polynomials and Quasimodular Forms*, arXiv:2602.10536, submitted
4. E. Chen, K. Lau, **S. Lee**, K. Ono, J. Zhang, *Dead ends in square-free digit walks*, arXiv:2602.05095
5. G. Bates, R. Jesubalan, **S. Lee**, J. Lu, H. Shim, *Powerful Fibonacci polynomials over finite fields*, arXiv:2601.02664, submitted
6. J. Getz, A. G. Terradillos, F. Hosseinijafari, B. Hu, **S. Lee**, A. Slipper, M.-H. Tomé, H. Yao, A. Zhao, *Modulation groups*, arXiv:2510.23932, submitted
7. J. Baek, **S. Lee**, *Formalizing Mason–Stothers Theorem and its Corollaries in Lean 4*. arXiv:2408.15180
8. **S. Lee**, Y. Choi, J. Park, B. Kim, J. Shin, *Consistency and Monotonicity Regularization for Neural Knowledge Tracing*, arXiv:2105.00607
9. Y. Choi, Y. Lee, J. Cho, J. Baek, D. Shin, H. Yu, Y. Shim, **S. Lee**, J. Shin, C. Bae, B. Kim, J. Heo, *Assessment Modeling: Fundamental Pre-training Tasks for Interactive Educational Systems*, arXiv:2022.05505

## Awards, Grants & Honours

|                                                                                        |                     |
|----------------------------------------------------------------------------------------|---------------------|
| Department of Mathematics Summer Grant, UC Berkeley . . . . .                          | 2024 Summer         |
| Outstanding Graduate Student Instructor Award, UC Berkeley . . . . .                   | 2024 Spring         |
| Graduate Student Researcher, UC Berkeley . . . . .                                     | 2023 Spring, Summer |
| Kwanjeong Educational Foundation Scholarship, KEF . . . . .                            | 2017–2018           |
| Excellency Award (Top Honours), Dept. of Mathematics, POSTECH . . . . .                | 2017                |
| POSTECH Outstanding Talent Development Scholarship, POSTECH . . . . .                  | 2013–2016           |
| National Science and Technology Scholarship, KOSAF . . . . .                           | 2013–2016           |
| Silver medals, Undergraduate Mathematical Competition, KMS . . . . .                   | 2013, 2015, 2016    |
| 31st place, ACM-ICPC Daejeon Regional, ACM . . . . .                                   | 2015                |
| Grand prize, POSTECH Programming Contest, Dept. of Computer Science, POSTECH . . . . . | 2015                |
| Honorable mention, Korean Olympiad of Informatics, NIA . . . . .                       | 2012                |

## Teaching Experience

### Graduate Student Instructor (T.A.)

UC Berkeley

Berkeley

2019 – Present

- (2026 Spring) Multivariable Calculus
- (2025 Fall) Introduction to Abstract Algebra
- (2025 Spring) Cryptography
- (2025 Spring) Introduction to Mathematical Logic

- (2024 Fall) Abstract Linear Algebra
- (2024 Spring) Methods of Mathematics: Calculus, Statistics, and Combinatorics
- (2023 Fall) Methods of Mathematics: Calculus, Statistics, and Combinatorics
- (2022 Fall) Multivariable Calculus
- (2019 Spring) Methods of Mathematics: Calculus, Statistics, and Combinatorics

#### **Grader & T.A.**

POSTECH

Pohang  
2016 – 2018

- (2018 Spring) Differential Manifolds and Lie groups (Graduate course)
- (2017 Fall) Modern Algebra II
- (2017 Spring) Calculus
- (2016 Fall) Applied Linear Algebra (Undergraduate T.A.)

#### **Tutoring**

POSTECH

Pohang  
2014 – 2015

- (2015 Spring) Calculus
- (2015 Spring) Modern Algebra I
- (2014 Fall) Analysis II
- (2014 Spring) Analysis I

### **Outreach**

#### **KIAS Winter School on Mathematics and AI**

Team Leader

Jeongseon  
2025 Winter

- Use machine learning to study Artin representations and abelian varieties over finite fields
- Team members: Dohoon Choi, Keunyoung Jeong, Jaehak Yi, Minseo Shin, Euntaek Lee, Taeyoung Kim

#### **Berkeley Math REU**

Mentor

Berkeley  
2025 Summer

- Diophantine equations on Fibonacci polynomials and ties in Chebyshev bias over finite fields
- Mentees: Graeme Bates, Ryan Jesubalan, Jane Lu, Hyewon Shim

#### **Directed Reading Program**

Mentor

Berkeley  
2023-present

- (2025 Fall) Group cohomology (Graeme Bates, Jane Lu)
- (2025 Spring, 2025 Fall) Modular forms (Dongho Kim)
- (2023 Fall) Elliptic curves (Jacob Martin)
- (2023 Spring)  $p$ -adic numbers (Lucas Xie)

#### **POSTECH Potential Development Camp for High School Students**

Mentor

POSTECH  
2015 Winter

- Mentoring high school students for college-level mathematics

### **Talks**

- Research Talks

- Mathematics Colloquium, Carnegie Mellon University, Pittsburgh, April 2026  
Positive Quasimodular Forms and Linear Programming Bounds
- Aarhus Mathamtics & AI Workshop, Aarhus Institute of Advanced Studies, Aarhus, January 2026  
How Machines Learn Galois Groups
- Lean Together 2026, Online, January 2026  
Progress on the Sphere Packing Project (with Sidarth Hariharan and Chris Birkbeck)
- June E Huh Center for Mathematical Challenges, KIAS, Seoul, January 2026  
Positive Quasimodular Forms and Linear Programming Bounds
- Number Theory Seminar, University of Wisconsin-Madison, Wisconsin, December 2025  
How Machines Learn Galois Groups
- EPFL, Lausanne, November 2025  
Sphere Packing, Sign Uncertainty Principle, Universal Optimality, and Positive Quasimodular Forms
- Aarhus University, Aarhus, November 2025  
Sphere Packing, Sign Uncertainty Principle, Universal Optimality, and Positive Quasimodular Forms
- Special Seminar, Carnegie Mellon University, Pittsburgh, November 2025  
How Machines Learn Galois Groups
- International Seminar on Automorphic Forms, Online, April 2025  
Algebraic proof of modular form inequalities for optimal sphere packings
- RTG seminar, UC Berkeley, Berkeley, February 2025  
Algebraic proof of modular form inequalities for optimal sphere packings
- Algebra Discrete Math seminar, UC Davis, Davis, January 2025  
Algebraic proof of modular form inequalities for optimal sphere packings
- 6th EU/US Workshop on Automorphic Forms and Related Topics, Luminy, September 2024  
Algebraic proof of modular form inequalities for optimal sphere packings
- POSTECH Number Theory Seminar, POSTECH, Pohang, May 2024  
Algebraic proof of modular form inequalities for optimal sphere packings
- Student Number Theory Seminar, UC Berkeley, Berkeley, April 2024  
Algebraic proof of Viazovska's inequalities
- School of Mathematics, KIAS, Seoul, December 2023  
A new proof of Viazovska's modular form inequality and beyond
- International Conference on Machine Learning, Hawaii, US, July 2023  
HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- Center for Artificial Intelligence and Natural Sciences, KIAS, Seoul, June 2023  
HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- School of Computing, KAIST, Daejeon, June 2023  
HETAL: Efficient Privacy-preserving Transfer Learning with Homomorphic Encryption
- 1st FHE.org workshop, Trondheim, May 2022  
Encrypted Multinomial Logistic Regression Training with Softmax Approximation
- Workshop for Young Mathematicians in Korea, Online, January 2022  
Hitchhiker's guide to non-archimedean world
- Graduate student seminar, Sogang University, Seoul, July 2018  
Maass wave forms, quantum modular forms and Hecke operators
- Sungkyunkwan University, Seoul, June 2018  
Maass wave forms, quantum modular forms and Hecke operators

- NCTS-POSTECH Number Theory Workshop, NTU, Taiwan, December 2017  
Quantum modular forms and Hecke operators
- Expository Talks
  - Student Number Theory Seminar, Berkeley, March 2026  
Mathematics, AI, and Formalization
  - Harmonic Analysis Seminar, University of California Irvine, February 2026  
Mathematics, AI, and Formalization
  - Student Number Theory Seminar, Berkeley, August 2025  
How Do Automorphic Forms and Elliptic Curves Fly? (Survey on Murmuration)
  - Bruhat-Tits building seminar, Berkeley, February 2025  
Bruhat-Tits buildings for split groups / Moy-Prasad filtration and Local Langlands correspondence
  - Berkeley–Stanford Number Theory Learning Seminar, Berkeley, December 2024.  
Proof of irrationality of  $L(2, \chi_{-3})$  and product of log values
  - Student Number Theory Seminar, Berkeley, October 2024  
Modular forms on  $G_2$
  - Geometric class field theory learning seminar, Berkeley, Sep 2024  
Singular algebraic curves and de-normalization
  - Student Number Theory Seminar, Berkeley, March 2024  
Linear Programming Beyond Sphere Packing
  - Orbit methods and automorphic forms learning seminar, Berkeley, Oct 2023  
Gan–Gross–Prasad conjectures
  - Student Number Theory Seminar, Berkeley, Nov 2022  
Shimura correspondence and Waldspurger’s formula
  - Instructional Workshop on Class Field Theory, KIAS, Seoul, January 2018  
Proof of the main theorem of local class field theory

## Languages

- Korean (native), English (fluent)
- Python (PyTorch, Numpy, Pandas), C/C++,  $\text{\LaTeX}$ , SAGE Math, Lean, MATLAB, Haskell

## Miscellaneous (click the icons)

- Working as a reviewer for Mathematical Reviews (2022~) 
- GitHub blog on various topics 
- Math Stackexchange  & Math Overflow 
- Speedcuber 
- DJ (Techno, Trance, House)  

(Last updated: May 16, 2026)